



with eight or twelve nozzles, Europlacer's ii A1 model is said to deliver flexibility in handling a broad range of components with 164 feeder positions and two internal matrix trays, all in a compact footprint. It also features the firm's patented intelligent conveyor with board stops under full software control.

Software suite. If you are considering purchasing automatic machines, how are they programmed and what are the software used should be answered. The software has three primary goals of easy use, flexibility, and optimising performance.

Whatever, remember to check if your model comes with a computer or just the software. This can neither be good nor bad as some users prefer to install the software on their own PCs. A fully integrated system is the best choice to avoid software compatibility issues, however.

Also check if the machine runs on a familiar graphical user interface (GUI), such as Windows, or a proprietary system. Most operators prefer a standard GUI since a proprietary GUI may require a longer learning curve. CAD download, teaching camera, barcode readers, management information system (MIS), and optimisation functions and off-line program-

ming can all make the operator's job easy and efficient. Hence, a strong software suite is important to make the process easy and traceable.

For instance, DDM Novastar adds a unique programming software—SMTrue Run Optimise—to the full-scale SMT assembly services it offers. The software is used for making offline programming of automatic SMT machines to derive increased efficiency and throughput.

ASM Assembly System also has IIoT interfaces for integration into line and factory systems (ASM OIB, IPC-Hermes 9852, IPC-CFX) and the cloud (ADAMOS) with advanced workflow solutions for setup processes, material management, factory monitoring, and factory integration.

The new machines are also using a lot of open source software. OpenPnP is an open source SMT pick and place system that includes ready-to-run software and hardware designs that you can build and modify. Index PnP is an open source pick-and-place machine that is among the recent launches designed for mass-production volumes typical of crowdfunding projects.

Other important considerations

Populating more boards and machines that are fit for any intelligent factory, complete with fully automated material flow, faster changeovers and open interface software



Mycronic MY300 (Credit: www.mycronic.com/en)

is the real pursuit of the industry. There is no limit to the minutiae of pick-and-place machine specifications. You may have to select the features as per your needs.

Unlike the rapid placers, precision placers often do not use mounted nozzles. They may rely on gantry-supported moving heads with relatively few pickup nozzles. While the conventional four-head vertical heads were only available in table-top machines, now they have evolved into automated machines.

Some machines may have to optionally be fitted with a dispense head for depositing adhesive on the board. If you require this feature, consider the dispense method, dot size, and dispense speed.

Maximum and minimum PCB or panel size and thickness can be critical requirements for some manufacturers. Fiducial recognition, coordinate correction, and bad mark detection should be standard for any automatic PnP machine.

New machines have a 3D stereo measurement feature for the through-hole device that easily identifies the image of the vertical pin and height of the component lead to detect bending deformation. This tells whether the through-hole components are placed accurately to ensure the correct height after placement, and thus reduces the need for operator's visual inspection. Simi-



Europlacer's ii A1 model (Credit: <https://europlacer.com>)